

T01 THERMAL PERFORMANCE | P

Intent : Provide a thermal environment that the majority of building users find acceptable.

Summary : This WELL feature requires projects to create indoor thermal environments that provide comfortable thermal conditions to the majority of people in support of their health, well-being and productivity.^{1,2}

Issue : The thermal environment substantially impacts a building's energy footprint, since in many countries heating and cooling account for approximately half of a building's energy consumption.^{3,4} Furthermore, the indoor thermal environment is ranked as one of the strongest contributing factors to overall human satisfaction in the built environment.⁵ The thermal environment not only impacts comfort and productivity, but due to its linkages to integumentary, endocrine and respiratory body systems, thermal comfort can also cause a variety of detrimental health outcomes.^{1,6} Overly warm indoor spaces are linked to increases in sick building syndrome symptoms, irregular heart rate, respiratory issues, fatigue and negative mood.⁷ Cold work environments have been linked to increased effort and work towards maintaining proper posture and increased risk for chronic issues related to musculoskeletal health.⁸

Solutions : Thermal comfort standards utilize a model that provides a means of predicting whether humans in a mechanically conditioned space will be satisfied with the thermal environment based on six core parameters: air temperature, humidity, air movement, mean radiant temperature of surrounding surfaces, metabolic rate and clothing insulation.^{9,10} For naturally conditioned buildings, the adaptive thermal comfort model correlates human comfort directly with indoor operative temperature and outdoor temperature.^{11,12} Achieving thermal satisfaction among people requires some level of control over thermal comfort parameters in any given environment.

PART 1 PROVIDE ACCEPTABLE THERMAL ENVIRONMENT

For All Spaces except Commercial Kitchen Spaces:

Option 1: Performance verified environmental conditions

The following requirements are met, as applicable:

- Mechanically conditioned regularly occupied spaces maintain thermal comfort conditions of PMV +/- 0.5 for at least 90% of occupied hours in at least 90% of regularly occupied spaces.¹³
- Naturally conditioned regularly occupied spaces meet all the following conditions.¹²

	Prevailing Mean Outdoor Temperature, $t_{pma(out)}$	Indoor Operative Temperature	Notes
Minimum	10 °C	$t_{pma(out)} \times 0.31 + 14.3$ °C	N/A
Maximum	33.5 °C	$t_{pma(out)} \times 0.31 + 21.3$ °C	Occupant-controlled elevated air speed may be used to increase this maximum per ASHRAE 55

- Mixed-mode-conditioned spaces meet the requirements for both mechanically and naturally conditioned spaces, when each is in operation

Note : Spaces with multiple occupant types may use the occupant characteristics of PMV—metabolic rate and clothing value—for the majority occupant type (e.g., a space with 30 diners and 5 servers may use the diners' metabolic rates and clothing values to determine the PMV for that space).

OR

Option 2: Long-term thermal data

The following requirements are met:

- Project meets Feature T06 Thermal Comfort Monitoring.
- Sensor data demonstrate parameters meet one of the following:
 - One of the PMV or temperature ranges described in Option 1. Dry bulb temperature may be used in place of operative temperature. Naturally conditioned projects must also measure outdoor air temperature.
 - Dry bulb temperature is between 21-25 °C for occupied hours.¹⁴ The designed air velocity is not more than 0.2 m/s at 1.7 m above the floor.

Note : Spaces with multiple occupant types may use the occupant characteristics of PMV—metabolic rate and clothing value—for the majority occupant type (e.g., a space with 30 diners and 5 servers may use the diners' metabolic rates and clothing values to determine the PMV for that space). Projects undergoing recertification which were previously awarded Feature T06 must consider all data collected since the previous certification.

OR

Option 3: Thermal comfort surveys

The following requirement is met:

- The project achieves at least two points in Feature T02: Verified Thermal Comfort.

For Commercial Kitchen Spaces:

The following requirement is met:

- The operative temperature in the kitchen does not exceed 27 °C.

Note :

Multifamily residential projects may achieve WELL Certification at the Bronze or Silver level without testing in dwelling units, but cannot achieve Gold or Platinum without testing in dwelling units. See Sampling Rates for Multifamily

Residential in the WELL Performance Verification Guidebook for further details.

Refer to the Performance Verification Guidebook for information on sensor/testing requirements, required testing duration and compliance calculations.

PART 2 MEASURE THERMAL PARAMETERS

For All Spaces except Dwelling Units:

Option 1: Semi-annual testing

The following requirements are met:

- a. The dry-bulb temperature, relative humidity, air speed (only for projects that use elevated air speed method) and mean radiant temperature are measured in regularly occupied spaces at intervals no longer than twice per year (including once in June, July or August and once in December, January or February), and the results are submitted annually through the WELL digital platform.
- b. The number and location of sampling points comply with the requirements outlined in the Performance Verification Guidebook.

Note: Projects are not required to follow the device requirements or test methods described in the Performance Verification Guidebook. Permanently installed monitors may be located on interior walls rather than a minimum distance away.

OR

Option 2: Continuous monitoring

The following requirement is met:

- a. Project meets Feature T06 Thermal Comfort Monitoring.

REFERENCES

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